

Dual-branch bidirectional attention fusion of temporal and hierarchical representations for robust ECG arrhythmia classification

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ARTICLE INFO

Keywords:

ECG
Arrhythmia classification
Bidirectional attention fusion
Capsule network
Class imbalance
Domain shift

ABSTRACT

Electrocardiography (ECG) is a core biomedical signal for arrhythmia assessment, yet automated analysis remains challenged by acquisition noise, inter-patient variability, and severe class imbalance. Existing deep models often emphasize either temporal dynamics or hierarchical morphology, while many dual-branch designs still rely on static or one-way fusion that only weakly captures cross-branch dependencies. This study proposes a dual-branch framework for robust ECG arrhythmia classification that integrates a multi-scale CNN-SE-BiLSTM branch for temporal representation learning with a Capsule Network branch enhanced by multi-head self-attention for hierarchical morphology modeling. The two branches are fused by an Adaptive Multi-head Bi-Directional Attention Fusion (AMBAF) module that performs sample-specific, beat-wise cross-branch reweighting. To improve the minority-class discrimination, an Enhanced Adaptive Margin Context Loss (EAMCL) introduces class-aware adaptive margins without relying on oversampling. On the MIT-BIH Arrhythmia Database under the AAMI five-class inter-patient protocol, the proposed model achieves 99.55% accuracy with strong precision, recall, and F1-score across classes. Cross-dataset transfer on PTB, INCART, and SVDB further supports robustness under domain shift and acquisition variability.

1. Introduction

Cardiovascular diseases remain a major cause of mortality worldwide, creating sustained demand for reliable automated tools that assist ECG interpretation. As a non-invasive biosignal, the electrocardiogram (ECG) supports arrhythmia screening and cardiac assessment, but its analysis is complicated by acquisition noise, inter-patient variability, waveform heterogeneity, and the need for expert interpretation. These challenges make robust ECG classification an important problem in biomedical engineering and computer-aided diagnosis.

Existing AI-based ECG recognition methods range from traditional

machine learning pipelines to end-to-end deep learning. Machine learning approaches typically follow a two-stage process: handcrafted feature extraction using signal processing methods followed by classification. For example, Discrete Wavelet Transform (DWT) (Shuvo et al., 2025) and Fast Fourier Transform (FFT) (Gupta et al., 2021) are commonly used to derive representative features, which are then input to classifiers such as Support Vector Machines (SVM) (Bhaskar et al., 2025), Random Forests (Alimam et al., 2025), and XGBoost (Pandey et al., 2021).

In contrast, deep learning methods operate directly on raw ECG signals and enable end-to-end learning, reducing manual feature

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engineering across time, frequency, and morphological domains (Fuadah and Lim, 2025). Convolutional Neural Networks (CNNs) learn waveform morphology and salient local patterns (e.g., P-wave, QRS complex, and T-wave), while Long Short-Term Memory (LSTM) networks model temporal dependencies that can improve arrhythmia recognition (Shrivastava et al., 2023). Capsule Networks (CapsNet), through capsule structures and routing mechanisms, encode hierarchical relationships among learned features and can improve robustness to waveform variations (Shah et al., 2023). Complementary mechanisms have also been explored to improve measurement reliability, including attention modules that emphasize informative segments (Guhdar et al., 2025), transformer-based architectures for long-range dependencies (Hu et al., 2022), and imbalance-mitigation strategies such as Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al., 2002).

From a measurement perspective, ECG signals are physiological time-series measurements whose interpretation should remain accurate and reproducible despite noise, inter-patient variability, and subtle pathological deviations. Therefore, robust ECG analytics should (i) capture both beat-level local morphology and rhythm-level temporal dependencies, (ii) preserve complementary information across representation levels rather than collapsing it through fixed fusion, and (iii) maintain reliable minority-class detection under the severe class imbalance typical of arrhythmia datasets (Al-Shannaq et al., 2025; Rath et al., 2021; Sharma et al., 2023; Febeena and Kurian, 2025; Bai et al., 2024; Wang, 2023). However, several recurring challenges persist. Single-model designs are often insufficient to simultaneously extract diverse features and achieve stable classification under varying acquisition conditions (Xu et al., 2024). Some dual-model strategies emphasize high-level representations while weakening low-level temporal cues that are critical for arrhythmia discrimination (Lilda and Jayaparvathy, 2025). Moreover, when multiple modules are combined, fusion is frequently implemented using simple concatenation/averaging, which may overlook important cross-level spatiotemporal correlations (Talukder, 2025). Many approaches also model temporal dependencies unidirectionally, ignoring future context that can be informative within a fixed ECG segment (Al-Shannaq et al., 2025). In addition, imbalance handling remains inadequate, leading to degraded minority-class performance (Rath et al., 2021). Finally, a number of studies prioritize headline accuracy without sufficient analysis of class-wise behavior, limiting diagnostic reliability and transparency in practical monitoring (Sharma et al., 2023).

To address these limitations, this work targets a central gap in ECG biosignal analysis: temporal and hierarchical features are frequently learned in isolation, while their cross-level dependencies are only weakly modeled during fusion. We therefore propose a dual-branch framework for robust ECG arrhythmia classification. The low-level branch combines multi-scale convolutions, squeeze-and-excitation, and BiLSTM layers to model fine-grained temporal dynamics and contextual rhythm information. The high-level branch employs a Capsule Network with multi-head self-attention to capture hierarchical waveform morphology and part-whole relationships. Their outputs are integrated by the Adaptive Multi-head Bi-Directional Attention Fusion (AMBAF) module, which enables sample-specific cross-branch interaction instead of fixed concatenation or averaging. To further strengthen minority-class recognition, the Enhanced Adaptive Margin Context Loss (EAMCL) introduces class-aware adaptive margins without oversampling-based augmentation. Accordingly, the objective of this study is to develop and validate a robust ECG classification framework that jointly models temporal and hierarchical representations under class imbalance, acquisition variability, and domain shift.

Specifically, the main contributions are:

1. A dual-branch ECG representation framework that jointly models low-level temporal dynamics through a multi-scale CNN-SE-BiLSTM branch and high-level hierarchical morphology through a Capsule Network with multi-head self-attention.

2. An Adaptive Multi-head Bi-Directional Attention Fusion (AMBAF) module that performs dynamic cross-branch interaction and sample-specific reweighting, enabling explicit modeling of temporal-hierarchical dependencies beyond static fusion.
3. An Enhanced Adaptive Margin Context Loss (EAMCL) that introduces class-frequency-dependent margins to mitigate class imbalance and improve minority-class separability.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the proposed method; Section 4 describes the experimental setup; Section 5 reports the results and comparative analysis; Section 6 concludes and outlines future work.

2. Related work

ECG recordings are biomedical time-series signals whose clinical value depends on reliable analysis under variable acquisition conditions. Recent ECG classification research has increasingly focused on deep representation learning, fusion of complementary features, and evaluation protocols that account for class imbalance, noise, and domain shift. The following review highlights these themes and identifies the remaining methodological gap addressed in this study.

2.1. Capturing temporal dynamics and multi-scale patterns

A common direction is combining CNN-based feature extraction with recurrent networks for temporal modeling. Febeena et al. (Febeena and Kurian, 2025) proposed C-BiNET, a CNN-BiLSTM design that combines wavelet-based denoising with ADASYN augmentation to strengthen temporal modeling under class imbalance. Bai et al. (Bai et al., 2024) integrated CNNs with Bidirectional Gated Recurrent Units (BiGRU) and attention, demonstrating strong generalization on multi-lead signals. Wang et al. (Wang, 2023) introduced a CNN-BiGRU design with Spatial and Channel Squeeze-Excitation modules to recalibrate features. These approaches improve temporal modeling, but fusion is often shallow or effectively one-directional, which can limit cross-level interaction between representations.

2.2. Modeling global context using attention and Transformers

Attention-based models aim to capture long-range dependencies that are difficult to model with standard CNN/RNN pipelines. Islam et al. (Islam et al., 2023) presented a convolutional network with heartbeat-aware attention to address class imbalance. Wang et al. (Wang et al., 2023) developed ACA-MA using multi-head self-attention mechanism to capture global context. Akan et al. (Akan et al., 2023) introduced ECGformer for arrhythmia classification, while EL-Ghaish et al. (EL-Ghaish and Eldele, 2024) proposed a Bidirectional Transformer framework. Zhou et al. (Zhou and Wang, 2024) combined multi-branch CNNs and Transformer architectures with RR-interval features to improve robustness across protocols. Although powerful, such architectures increase computational cost and still do not explicitly separate low-level temporal and high-level hierarchical feature learning.

2.3. Multi-branch architectures and the fusion limitation

Multi-branch designs aim to extract complementary representations, but the fusion strategy is often the main bottleneck. Xu et al. (Xu et al., 2024) combined state-space modeling (Mamba) with Capsule Networks for interpretable outputs. Lilda et al. (Lilda and Jayaparvathy, 2025) introduced a multi-branch CNN operating on ECG spectrograms. Talukder et al. (Talukder, 2025) developed a hybrid multiscale ConvNeXt-based model. Pan et al. (Pan and Wu, 2025) combined CNNs with Hidden Markov Models (HMMs) to improve efficiency for deployment scenarios. These studies support the value of multi-representation learning, yet many still rely on simple concatenation,

averaging, or limited interaction, which can discard important complementary information and weaken cross-branch dependency modeling.

2.4. Class imbalance mitigation and reliability concerns

Because ECG datasets are often heavily imbalanced, many studies apply explicit mitigation strategies. Rafi et al. (Rafi and Ko, 2022) and Xia et al. (Xia et al., 2023) use Generative Adversarial Networks (GANs) to synthesize minority-class samples. C-BiNET (Febeena and Kurian, 2025) addresses imbalance through ADASYN-based adaptive synthetic sampling as a data augmentation step, while STL-HMFF (Talukder, 2025) uses SMOTE-TomekLink with oversampling. However, oversampling can duplicate patterns, increase overfitting risk, and obscure true generalization under realistic clinical class distributions.

2.5. Noise sensitivity and domain shift evaluation

Noise remains an important reliability challenge in ECG classification, as ECG recordings are often affected by acquisition-related disturbances that can distort clinically relevant waveform patterns. ECG denoising methods can be grouped into three categories. Classical ECG denoising methods are mainly signal-processing based and usually depend on fixed assumptions or parameter tuning, such as Wavelet Transform (Chen and Tsui, 2020) and DWT-based filtering (Jenkal et al., 2024). Data-driven ECG denoising methods learn noise-removal patterns directly from data, such as DRNN (Antczak, 1807) and CGAN (Wang et al., 2022). Advanced generative and diffusion-based denoising methods progressively reconstruct clean ECG signals from noisy observations, such as DeScod-ECG (Li et al., 2023) and the IDPM-based model (Hassoon et al., 2025).

The Arrhythmia literature has established that high performance on a single database does not necessarily translate to reliable deployment across patients, devices, leads, or clinical sites. For this reason, many studies have tried to reduce this gap by aligning source and target feature distributions through various transfer learning protocols. Niu et al. (Niu et al., 2020) proposed an improved ECG arrhythmia classification across different patient domains by using a domain-adversarial neural network (DANN). Other studies used maximum mean discrepancy (MMD) minimization to directly reduce the distribution shift. Li et al. (Li et al., 2025) developed a domain generalization framework for cardiac signal analysis that uses MMD minimization to align feature distributions across multiple source datasets. Beyond the adversarial and discrepancy-based alignment, feature-normalization-based adaptation provides a lightweight way to reduce domain shift. Li et al. (Li et al., 2018) proposed AdaBN as modulating batch-normalization statistics across the network for domain adaptation.

These studies show that noise-aware analysis and domain-shift evaluation are essential for reliable ECG classification. Nevertheless, these aspects are often investigated separately from representation fusion and imbalance mitigation. This motivates our analysis to evaluate the efficacy of these aspects to the proposed framework.

2.6. Efficient alternatives and remaining gaps

Traditional machine learning methods can still be useful when computational complexity is a key constraint. Jayaparththy et al. (Jayaparththy, 2024) employed Extreme Gradient Boosting with Grey Wolf Optimization to classify cardiovascular diseases using fewer features. Recent studies also continue to emphasize robust ECG measurement analytics through multi-branch fusion or signal-adaptive feature extraction, including a dual-branch spatiotemporal design for multi-lead arrhythmia classification (Cai et al., 2025) and a wavelet-assisted BiLSTM pipeline for arrhythmia classification (Tchinda et al., 2025). However, a persistent gap remains: many multi-branch approaches either (i) do not fuse low-level temporal cues with high-level

hierarchical representations, or (ii) adopt fixed or shallow fusion that cannot perform beat-wise, cross-branch reweighting.

Distinct from prior hybrid ECG classification methods such as ECG-former (Akan et al., 2023) and MambaCapsule (Xu et al., 2024), which often fuse representations via concatenation or limited attention, the present work introduces an Adaptive Multi-head Bi-Directional Attention Fusion (AMBAF) mechanism that bidirectionally fuses low-level temporal and high-level hierarchical features through cross-branch interaction and sample-specific reweighting. In addition, the Enhanced Adaptive Margin Context Loss (EAMCL) addresses imbalance through adaptive, class-aware margins rather than relying on oversampling. Together, these design choices target both robust representation learning and reliable performance under clinical class imbalance, improving diagnostic usefulness and transparency.

3. Methodology

This section presents the main architectural components of the proposed framework, as illustrated in Fig. 1. The framework is organized into 3 key modules: 1) a low-level feature extraction branch based on an enhanced BiLSTM network, which captures fine-grained temporal and morphological ECG patterns; 2) a high-level feature extraction branch based on an improved capsule network, which models hierarchical waveform relationships and part whole feature dependencies; and (3) an adaptive fusion module, which integrates the complementary representations learned by both branches to support more discriminative ECG classification. To further improve learning under class imbalance, model optimization is guided by an Enhanced Adaptive Margin Context Loss (EAMCL), which is a customized objective function we proposed to strengthen classification performance on imbalanced ECG datasets.

3.1. Low-level feature extraction branch (BiLSTM)

A specific limitation of the standard BiLSTM technique in the analysis of ECG signals is that they suffer from feature degradation over long sequences, which may cause the network to lose highly discriminative local morphological patterns extracted by preceding convolutional layers. To address this problem, we introduce an improved BiLSTM network that integrates three key components shown on the left end Fig. 1: 1) a preceding feature extraction module, 2) squeeze-and-excitation (SE) recalibration stage, and 3) a dedicated residual connection spanning the recurrent layers. While the preceding feature extraction layer focuses on local waveform morphology, the BiLSTM module is responsible in capturing the contextual relationships across the temporal axis. This is important because the interpretation of an ECG segment may depend on both preceding and succeeding waveform patterns. The main advantage of our structural modifications is that the branch retains sharp local convolutional morphology through the residual connection while effectively incorporating bidirectional temporal context.

Feature extraction:

The feature extraction module is used to capture local ECG morphology at multiple temporal resolutions. Since ECG waveforms include both narrow events and wider morphological patterns, a single convolutional kernel is insufficient to capture all relevant local structures. Therefore, we applied three parallel one-dimensional convolutional layers with kernel sizes 3, 5, and 7 to the input sequence. The smaller kernel captures short and abrupt waveform variations and the larger kernels provide wider receptive fields for broader local patterns. The multi-scale convolutional features are computed as follows:

$$F_3 = \text{ReLU}(\text{Conv}_{k=3}(X)), F_5 = \text{ReLU}(\text{Conv}_{k=5}(X)), F_7 = \text{ReLU}(\text{Conv}_{k=7}(X)) \quad (1)$$

Each convolutional path in Eq. (1) produces 16 feature channels while preserving the temporal length through appropriate padding. The feature maps are then concatenated along the channel dimension, and

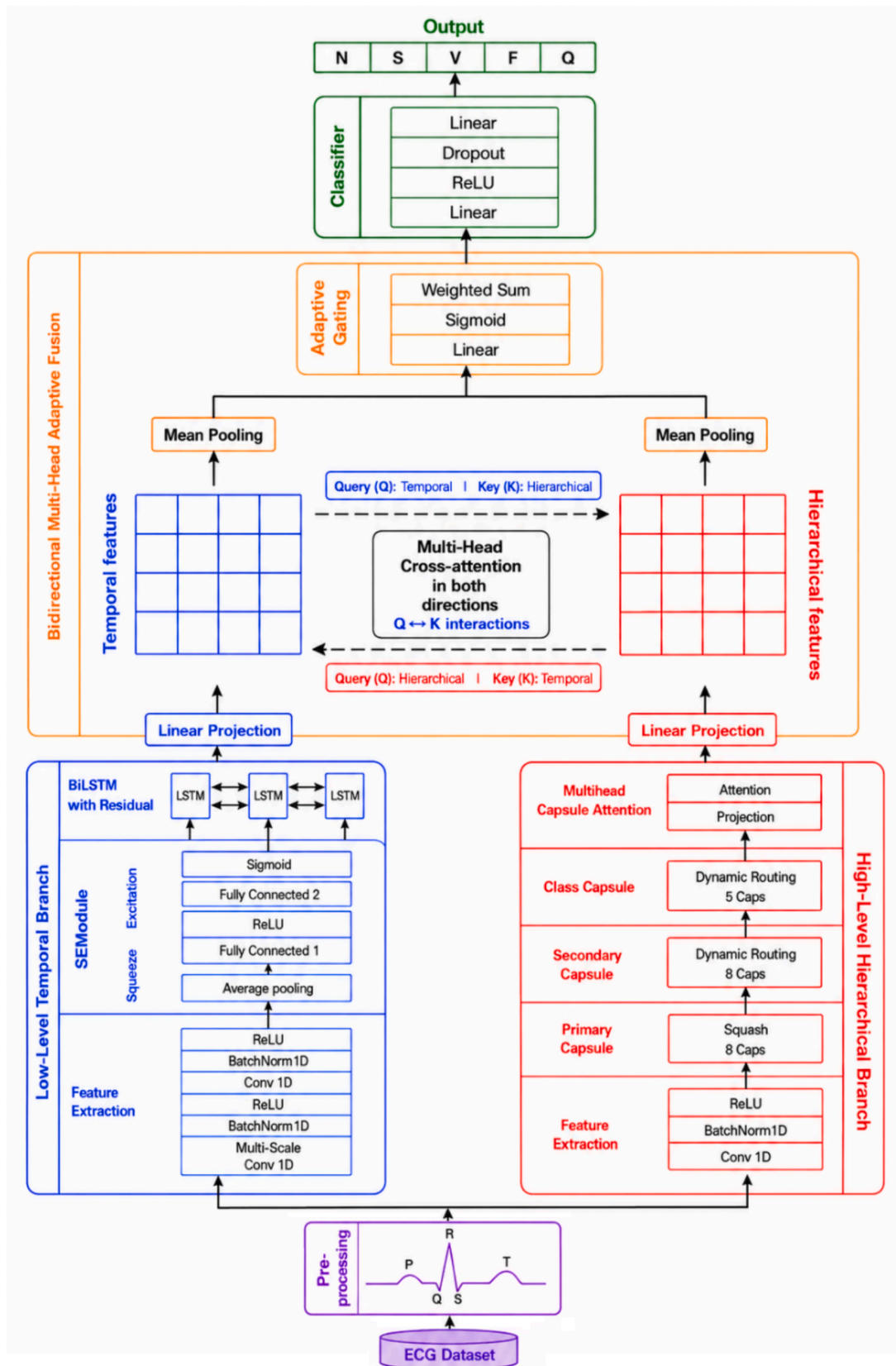


Fig. 1. The proposed dual branch framework.

batch normalization is applied, as shown in Eq. (2). The Batch normalization helps in stabilizing the distribution of intermediate features.

$$F^{bn} = \text{ReLU}(\text{BN}(\|F_3\|F_5\|F_7\|)) \quad (2)$$

A final one-dimensional convolution as shown in Eq. (3) is then used to integrate the concatenated multi-scale responses and reduce the channel dimension from 48 to 32. This operation produces a compact local representation that combines information from all 3 temporal receptive fields before attention-based recalibration.

$$F^e = \text{ReLU}(\text{BN}(\text{Conv}(F^{bn}))) \quad (3)$$

Squeeze-and-excitation attention:

The squeeze-and-excitation (SE) attention module is introduced to recalibrate the extracted convolutional channels according to their relative importance. The feature extraction module generates multiple local feature maps, but not all channels will contribute equally to the classification task. Some channels may capture discriminative waveform morphology, while others may respond to weak or redundant patterns. The SE module addresses this issue by learning channel-wise importance weights.

The squeeze operation summarizes each channel of the extracted feature map F^e using the global average pooling over the temporal dimension as shown in Eq. (4), while the excitation operation learns channel wise attention weights using two fully connected layers as shown in Eq. (5).

$$s_c = \frac{1}{T} \sum_{t=1}^T F_{t,c}^e, c = 1, \dots, C \quad (4)$$

$$s = \sigma(W_2 \times \text{ReLU}(W_1 \times z + b_1) + b_2) \quad (5)$$

It is important to note that $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$, where the values of reduction ratio r and the number of channels C are 4 and 32 respectively, resulting in an excitation path of $32 \rightarrow 8 \rightarrow 32$. The output attention vector is $s \in (0, 1)^C$ and the recalibrated feature map is obtained through channel-wise multiplication $\bar{F}_{t,c}^e = s_c \times F_{t,c}^e$. This recalibration preserves the temporal structure of the signal while emphasizing channels that contain more informative ECG morphology. Consequently, the subsequent BiLSTM module receives a more discriminative sequence representation.

Bidirectional long short term memory(BiLSTM) module:

The BiLSTM module is used to model bidirectional temporal dependencies from the recalibrated convolutional sequence. In our implementation, the recalibrated feature sequence $\bar{F}^e$ is passed to a two layer BiLSTM. The forward LSTM processes the sequence from 1 to T , while the backward LSTM processes the sequence from T to 1 as follows:

$$\vec{h}_t = \text{LSTM}_f(\bar{F}_t, \vec{h}_{t-1}), \bar{h}_t = \text{LSTM}_b(\bar{F}_t, \bar{h}_{t+1}) \quad (6)$$

The bidirectional hidden representation at timestep t is obtained by concatenating the two directional hidden states as shown in Eq (7), and the full BiLSTM output is shown in Eq (8).

$$h_t^b = [\vec{h}_t \| \bar{h}_t] \quad (7)$$

$$H^b = [h_1^b, h_2^b, \dots, h_T^b] \quad (8)$$

Each LSTM direction has its hidden size set to 32, resulting in a concatenated output hidden size of 64. To improve the gradient flow and preserve the local morphological information learned by the convolutional SE stages, a residual connection is added to the BiLSTM output. Since the SE output has dimension 32, while the BiLSTM output has dimension 64, the SE feature vector is expanded by channel-wise

duplication. This way, the residual enhanced BiLSTM representation is then computed as:

$$\bar{H}_t^b = h_t^b + R_t^b, t = 1, \dots, T \quad (9)$$

$$\bar{U}^b = \bar{H}^b \times W_b + b_b \text{ where } W_b \in \mathbb{R}^{T \times d} \quad (10)$$

This residual design allows the branch to retain local convolutional morphology while incorporating bidirectional temporal context from the BiLSTM. Finally, in Eq. (10) we linearly project the residual enhanced BiLSTM output into a shared embedding dimension d . The output preserves the temporal-token structure required by the bidirectional cross-attention fusion module.

3.2. High-level feature extraction branch (CapsNet)

Capsule Networks (CapsNet) are selected to model ECG morphology as a hierarchical composition of waveform primitives such as local deflections, QRS-related morphology, and surrounding beat-level patterns whose relative configuration is clinically meaningful. Unlike conventional CNN pipelines that may partially weaken spatial or morphological relational information through repeated pooling or fixed local aggregation, CapsNet represents features as vectors, where the vector length reflects the presence of a pattern and the vector orientation encodes instantiation attributes, while dynamic routing assigns lower-level parts to higher-level wholes based on agreement (Sabour et al., 2017). The classical dynamic routing captures agreement between lower-level and higher-level capsules, but with a known limitation, it does not explicitly model the pairwise, global dependencies among the final class-level capsule tokens. To overcome this design limitation, we propose an improved Capsule Network that integrates a multi-head self-attention mechanism directly applied over the output class capsules. This specific modification allows the network to explicitly refine inter-capsule relationships before the proposed fusion. The advantage of this structural improvement is the generation of a highly cohesive, globally-aware hierarchical representation. By capturing the mutual interactions between different morphological wholes, the network significantly improves discriminative power and robustness against inter-patient waveform variability.

This branch of the proposed framework on the right end of Fig. 1 targets the hierarchical morphological patterns of ECG signals, beginning with a convolutional layer followed by batch normalization. The output is then passed through three hierarchical capsule layers: primary, secondary, and class capsules. Each subsequent capsule layer uses dynamic routing to assign importance to the vectorized outputs of the preceding layer, thereby learning part-whole relationships. This routing mechanism computes the input to capsule j as:

$$s_j = \sum_i C_{ij} \hat{\mathcal{W}}_{ji} \quad (11)$$

where $\hat{\mathcal{W}}_{ji}$ is the prediction vector from capsule i to capsule j , and C_{ij} is the routing coupling coefficient learned during routing.

Each capsule output v_j is squashed using the following non-linear function:

$$u_j = \frac{\|s_j\|^2 s_j}{1 + \|s_j\|^2 \|s_j\| + \epsilon} \quad (12)$$

where $\|\cdot\|$ denotes the vector norm, and ϵ is a small constant for numerical stability.

Dynamic routing is performed using the routing-by-agreement mechanism shown in Fig. 2. After L routing iterations, the final CapsNet representation is obtained as:

$$U^C = [u_1, u_2, \dots, u_{N_c}] \in \mathbb{R}^{N_c \times d_{cap}} \quad (13)$$

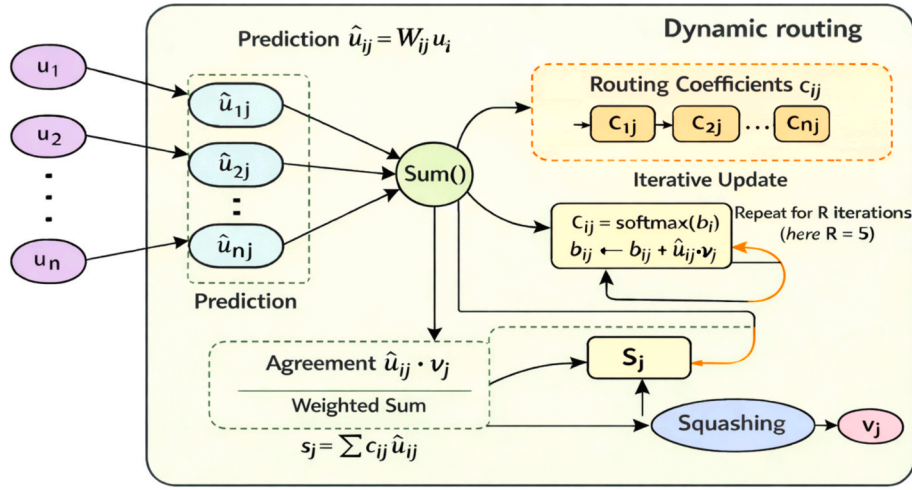


Fig. 2. Dynamic routing procedure in CapsNet.

where u_j is the output vector of the j -th capsule, N_c is the number of capsules in the output layer and d_{cap} is the dimensionality of each capsule's output vector. Multi-head self-attention is then applied over the capsule tokens to model interactions among the class-level representations, yielding an attention-refined capsule set. Although dynamic routing captures agreement between lower-level and higher-level capsules, it does not explicitly model pairwise dependencies among the final capsule tokens; therefore, multi-head self-attention is applied to refine inter-capsule relationships before fusion. This process involves computing separate query, key, and value projections for each attention head. Formally, for each head $m \in \{1, \dots, M\}$, the projections are obtained from the capsule matrix U^C as:

$$Q_m^C = U^C W_{m,Q}^C, K_m^C = U^C W_{m,K}^C, V_m^C = U^C W_{m,V}^C \quad (14)$$

where $W_{m,Q}^C, W_{m,K}^C \in \mathbb{R}^{d_{cap} \times d_k^C}$ and $W_{m,V}^C \in \mathbb{R}^{d_{cap} \times d_v^C}$. The capsule level self-attention matrix is then computed as Eq (15), which produces the output of the m -th capsule attention head O_m^C .

$$O_m^C = \text{softmax} \left(\frac{Q_m^C (K_m^C)^T}{\sqrt{d_k^C}} \right) \times V_m^C \quad (15)$$

The outputs of all capsule attention heads are concatenated in Eq (16) and projected back to the capsule dimension from the linear transformation in Eq (14). Lastly residual connection and layer normalization are applied to preserve the original routed capsule information, while incorporating capsule-token interactions. This operation is mathematically expressed in Eq (17)

$$MHSA^C = \text{Concat}(O_1^C, O_2^C, \dots, O_{n_c}^C) \times W_O^C \in \mathbb{R}^{N_c \times d_{cap}} \quad (16)$$

$$\bar{U}^C = \text{LayerNorm}(U^C + MHSA^C) \quad (17)$$

In all experiments, $M = 2$ heads are used. Finally, the attention-refined class capsules from $\bar{U}^C \in \mathbb{R}^{N_c \times d_{cap}}$ are forwarded to the fusion module (AMBAF) as a short capsule-token sequence.

3.3. Adaptive multi-head bi-directional attention fusion (AMBAF)

A multi-head bi-directional cross-attention fusion mechanism is introduced to integrate the outputs from the BiLSTM (low-level) and CapsNet (high-level) branches. Each branch is first projected to a shared embedding dimension, after which each branch attends to the other to explicitly model temporal-hierarchical correlations. A single-direction

attention mechanism would only condition one representation on the other, whereas the bi-directional design allows reciprocal refinement, enabling temporal tokens to be informed by capsule abstractions and capsule tokens to be informed by temporal context.

The AMBAF module performs cross-attention in two complementary directions between the projected BiLSTM ($\bar{U}^b \in \mathbb{R}^{T \times d}$) and CapsNet ($\bar{U}^c \in \mathbb{R}^{N_c \times d}$) branches, where $d = d_{bilstm} = d_{caps}$, T is the temporal sequence length and N_c is the number of output capsule tokens. For compactness purpose, the two attention directions are represented by the ordered pairs in Eq (18):

$$(X, Y) \in \{(B, C), (C, B)\} \quad (18)$$

where X denotes the query branch and Y denotes the context branch that provides the keys and values. The branch length is defined as $\dim(B) = T, \dim(C) = N_c$. $B \leftarrow C$ denotes the direction in which the BiLSTM branch queries the CapsNet branch, while $C \leftarrow B$ denotes the reverse direction in which the CapsNet branch queries the BiLSTM branch. For the m -th attention head, the query, key, and value projections for a given direction $X \leftarrow Y$ are computed as follows:

$$Q_m^{X \leftarrow Y} = U^X W_{m,Q}^{X \leftarrow Y} \in \mathbb{R}^{\dim(X) \times d_k} \quad (19)$$

$$K_m^{X \leftarrow Y} = U^Y W_{m,K}^{X \leftarrow Y} \in \mathbb{R}^{\dim(Y) \times d_k} \quad (20)$$

$$V_m^{X \leftarrow Y} = U^Y W_{m,V}^{X \leftarrow Y} \in \mathbb{R}^{\dim(Y) \times d_v} \quad (21)$$

where $W_{m,Q}^{X \leftarrow Y}, W_{m,K}^{X \leftarrow Y} \in \mathbb{R}^{d \times d_k}$ and $W_{m,V}^{X \leftarrow Y} \in \mathbb{R}^{d \times d_v}$. Here d_k and d_v computed as $d_k = d_v = d/n_h$ denote the key/query and value dimensions of each head respectively and n_h is the number of attention heads. The directional attention matrix is then computed using the scaled dot product attention equation in Eq. (22) and the output of the m -th attention head is computed using Eq. (23).

$$A_m^{X \leftarrow Y} = \text{softmax} \left(\frac{Q_m^{X \leftarrow Y} (K_m^{X \leftarrow Y})^T}{\sqrt{d_k}} \right) \quad (22)$$

$$O_m^{X \leftarrow Y} = A_m^{X \leftarrow Y} \times V_m^{X \leftarrow Y} \quad (23)$$

Based on our compactness definition, when $X = B$ and $Y = C$, the attention output has size $O_m^{B \leftarrow C} \in \mathbb{R}^{T \times d}$ which means that each temporal token is enriched using capsule-level information. On the other hand, $X = C$ and $Y = B$ the attention output has size $O_m^{C \leftarrow B} \in \mathbb{R}^{N_c \times d}$ meaning that each capsule token is refined using temporal contextual evidence. To get the final multi-head cross-attention outputs for each direction,

the outputs of all attention heads are concatenated and projected back to the shared embedding dimension using Eq (24), then followed with the independent application of residual connections and layer normalization for each cross-attention direction as expressed in Eq (25):

$$MH^{X \leftarrow Y} = \text{Concat}(O_1^{X \leftarrow Y}, O_2^{X \leftarrow Y}, \dots, O_{n_h}^{X \leftarrow Y}) \times W_0^{X \leftarrow Y} \in \mathbb{R}^{\dim(X) \times d} \quad (24)$$

$$\bar{U}^{X \leftarrow Y} = \text{LayerNorm}(U^X + MH^{X \leftarrow Y}) \quad (25)$$

Accordingly, the bidirectional cross-attention module produces two refined representations in Eq (26) and Eq (27). Specifically Eq (26) is a capsule aware temporal representation and Eq (27) is a temporally aware capsule representation. In this way, the module enables mutual information exchange between low-level temporal $\bar{U}^{B \leftarrow C}$ features and high-level hierarchical abstractions $\bar{U}^{C \leftarrow B}$ before the final gated fusion stage.

$$\bar{U}^{B \leftarrow C} \in \mathbb{R}^{T \times d} \quad (26)$$

$$\bar{U}^{C \leftarrow B} \in \mathbb{R}^{N_c \times d} \quad (27)$$

Since token level interaction has already been performed by the bidirectional cross-attention module, global average pooling is applied only after cross-branch refinement to obtain fixed-length branch summaries for gated fusion. The two outputs have different token dimensions, namely T temporal tokens and N_c capsule tokens, they are first transformed into fixed length branch level representations through global average pooling given in Eq (28) and Eq (29):

$$\bar{\bar{U}}^{B \leftarrow C} = \frac{1}{T} \sum_{t=1}^T \bar{U}^{B \leftarrow C} \quad (28)$$

$$\bar{\bar{U}}^{C \leftarrow B} = \frac{1}{N_c} \sum_{j=1}^{N_c} \bar{U}^{C \leftarrow B} \quad (29)$$

These two pooled representations summarize complementary information from the two branches. To adaptively combine these two representations, a feature wise gating mechanism is introduced. The gate is computed as follows:

$$g = \sigma(W_g [\bar{\bar{U}}^{B \leftarrow C} \parallel \bar{\bar{U}}^{C \leftarrow B}] + b_g) \quad (30)$$

where $g \in (0, 1)^d$, $W_g \in \mathbb{R}^{d \times 2d}$, $b_g \in \mathbb{R}^d$, $\sigma(\bullet)$ denotes the sigmoid activation function, and $\parallel$ denotes vector concatenation. The final fused representation is then obtained in Eq (31). Because $g \in (0, 1)^d$, the fusion is performed at the feature-dimension level rather than using a single scalar weight for the entire branch.

$$F = g \odot \bar{\bar{U}}^{B \leftarrow C} + (1 - g) \odot \bar{\bar{U}}^{C \leftarrow B} \quad (31)$$

where $\odot$ denotes element-wise multiplication. Unlike simple concatenation or averaging, this adaptive gate allows the model to dynamically regulate the contribution of each branch at the feature level. Therefore, the final representation can emphasize low-level temporal evidence when sequential dependencies are more informative, while assigning greater importance to capsule-level abstraction when hierarchical or class-level relationships are more discriminative. To complete the prediction task, the fused representation F is passed through a task specific prediction head given in Eq (32), and the class prediction probabilities are obtained by applying a SoftMax function to the output logits.

$$F^{out} = W_2 \times \text{ReLU}(W_1 \times F + b_1) + b_2 \quad (32)$$

Overall, the final fusion stage enables the model to integrate complementary information from both branches in an adaptive and content dependent manner. The BiLSTM branch contributes fine-grained

sequential dynamics and bidirectional temporal dependencies, while the CapsNet branch contributes high-level hierarchical and capsule-token relationships. By combining these representations through feature-wise gating rather than static concatenation, the proposed fusion module provides a more flexible mechanism for balancing temporal and hierarchical evidence in sequential data classification.

3.4. Enhanced adaptive margin context loss (EAMCL)

In imbalanced classification tasks, especially in medical datasets like MIT-BIH arrhythmia where the class distribution is highly imbalanced, deep learning models tend to favor the majority classes, specifically the normal class, resulting in suboptimal performance at detecting rare events. The conventional approach to assigning class weights generally involves directly inverting class frequencies or normalizing them to allocate greater weight to minority classes. However, this may result in disproportionately large weights for infrequent classes, which can amplify noisy minority samples and increase the risk of unstable optimization or overfitting.

To overcome this limitation, EAMCL provides an improved mechanism to handle class imbalance by imposing a class-dependent margin that dynamically scales according to class frequencies. This approach encourages the model to achieve higher separation between the true class and the most competitive false class, especially for minority classes.

1. Cross-entropy component

For a given input sample with logit vector $z \in \mathbb{R}^C$ and true label $y \in [1, \dots, C]$, the standard cross-entropy loss is defined as:

$$CE(z, y) = -\log \left(\frac{e^{z_y}}{\sum_{j=1}^C e^{z_j}} \right) \quad (33)$$

This term penalizes incorrect predictions and is widely used for multi-class classification.

2. Adaptive margin calculation

To adaptively penalize misclassification, a class-dependent margin m_y is defined based on the frequency of each class in the training set. Specifically, m_y is computed as follows

$$m_y = \frac{m_{\max}}{(n_y)^\alpha} \quad (34)$$

where n_y is the number of samples of class y , α controls the sensitivity to rarity, and $m_{\max}$ caps the maximum possible margin. This formulation imposes a tailored separation requirement for each class, where the minority classes receive larger margins, forcing the model to maintain a wider gap between the true logit and the highest competing false logit. In our implementation, we determined the appropriate values for $m_{\max}$ and α via grid search on a validation set, optimizing for macro-F1 across classes. The search ranges are $m_{\max} \in [0.2, 0.8]$ and $\alpha \in [0.1, 1.0]$.

3. Margin loss component

The margin loss is designed to ensure that the difference between the true logit z_y and the maximum logit for any incorrect class, $\max_{j \neq y} z_j$, exceeds the adaptive margin m_y . This is formulated as:

$$L_{\text{margin}}(z, y) = \max \left\{ 0, m_y - \left(z_y - \max_{j \neq y} z_j \right) \right\}^2 \quad (35)$$

If the margin condition $z_y - \max_{j \neq y} z_j \geq m_y$ is satisfied, the margin loss is zero; otherwise, the loss increases quadratically.

4. Combined loss function

The final Enhanced Adaptive Margin Context Loss (EAMCL) is defined as a weighted sum of the cross-entropy loss and the margin loss:

$$L(z, y) = CE(z, y) + \lambda * L_{margin}(z, y) \quad (36)$$

where λ denotes the balancing coefficient controlling the relative strength of the margin penalty. Across all experiments, λ is set to 0.1, selected on the validation split to maximize macro-F1. While Eq. (36) permits any nonnegative λ , the implementation restricts $\lambda \in [0, 1]$ to ensure stable training dynamics and avoid margin term dominating optimization.

By enforcing an adaptive margin, EAMCL forces the model to obtain a higher separation between the true class and the most competitive false class for minority classes. This targeted regularization improves the model's ability to correctly classify rare arrhythmia events, leading to more balanced class-wise performance, particularly for minority arrhythmia categories.

4. Experiment setup

This section presents the experimental framework for evaluating the proposed ECG classification approach. It includes the datasets used, the steps taken to prepare the data, details about how the model was implemented, and the metrics selected for evaluation, creating a clear basis for thoroughly assessing performance and comparing it with current standards in ECG classification.

4.1. Datasets

The MIT-BIH Arrhythmia Database serves as a cornerstone for evaluating the proposed ECG classification framework, renowned for its comprehensive arrhythmia annotations (Moody and Mark, 2001). It includes 48 half-hour, two-channel ECG recordings from 47 subjects, sampled at 360 Hz, with over 109,000 annotated heartbeats categorized into five classes: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), Cardiac Fusion beats (F), and Unknown (Q). In this context, F refers to beats with mixed morphology arising from the interaction of normal and ventricular activation, and it should not be confused with the fusion mechanism used in the proposed model architecture. The dataset's variety of arrhythmia patterns and pronounced class imbalance, especially for S and F beats, make it well-suited to test the EAMCL loss function, which targets skewed distributions. The signals were preprocessed to eliminate noise and baseline wander, segmenting them into heartbeats to enable robust assessment of the dual-branch model's performance across diverse clinical conditions.

The PTB Diagnostic ECG Database complements the evaluation by offering a broader scope for validating generalizability (Bousseljot et al., 1995). Comprising 549 records from 290 subjects, it includes healthy controls and multiple cardiac conditions recorded with 15-lead ECG (12 standard leads plus 3 Frank leads) sampled at 1000 Hz. For consistency with the proposed beat-level formulation, a single-lead setting was adopted and signals were resampled and segmented into heartbeat windows prior to transfer evaluation.

To extend cross-dataset validation beyond PTB, the St Petersburg INCART 12-lead Arrhythmia Database (INCART) was used as an additional target benchmark for transfer learning (Yakushenko, 2008). INCART contains 75 half-hour 12-lead Holter recordings from 32 subjects sampled at 257 Hz, with beat-level annotations.

Additionally, the MIT-BIH Supraventricular Arrhythmia Database (SVDB) was included to test robustness on supraventricular-rich recordings (Mark et al., 1990). SVDB provides 78 half-hour two-channel recordings sampled at 128 Hz, with beat annotations.

4.2. Datasets preprocessing

Effective preprocessing of ECG signals is crucial for robust

classification, particularly given the focus on cardiac beat extraction, as highlighted in prior work (Kachuee et al., 2018). The preprocessing pipeline consists of the following steps:

1. **Lead harmonization and resampling:** To ensure comparability across datasets with different lead configurations, and native sampling rates, a single-lead setting and a unified 360 Hz sampling rate are adopted. Specifically, MIT-BIH uses MLII (native 360 Hz), PTB uses Lead II (resampled from 1000 Hz), INCART uses Lead II (resampled from 257 Hz), and SVDB uses ECG1 (resampled from 128 Hz). Resampling follows a zero-phase (filtfilt) low-pass anti-aliasing filter.
2. **ECG Signal Segmentation:** The ECG signals are segmented into discrete 10-second windows, with a specific 10-second window selected for further processing to ensure consistency across the dataset.
3. **Amplitude Standardization:** To normalize the amplitude, the signal values are adjusted to fall within the range of zero to one, ensuring uniformity in signal representation across all samples.
4. **Local Maxima Detection:** The local maxima in each ECG segment are identified by detecting zero-crossings in the signal's first derivative, providing key reference points for further analysis.
5. **R-Peak Candidate Identification:** R-peak candidates are determined through a thresholding method, where a threshold of 0.9 times the local maxima is applied, a critical step for accurate heartbeat detection.
6. **Normal Heartbeat Period Estimation:** The typical value of the R-R interval is calculated by measuring the duration between consecutive R-peaks within a selected 10-second window, establishing a baseline for normal cardiac rhythm.
7. **Signal Length Equalization:** Each identified heartbeat, defined by R-peak intervals (1.2 times the R-R period), is segmented to ensure equal length across all samples, facilitating consistent input for the model.
8. **Zero Padding Adjustment:** Each selected signal segment is zero-padded to achieve a uniform, predefined length, ensuring compatibility with the model's input requirements and maintaining data integrity.

Class imbalance present in the ECG datasets was explicitly managed through the EAMCL loss function, which dynamically adjusts class-specific margins based on the frequency of each class. To clarify the extent of imbalance, Table 1 reports the number of beats per class after preprocessing and label mapping. For MIT-BIH, SVDB, and INCART, class counts are reported according to the AAMI beat-level classes N, S, V, F, and Q. Because the PTB Diagnostic ECG Database is organized for diagnostic cardiac-condition assessment rather than native AAMI beat-level arrhythmia annotation, its class distribution is reported according to the normal/abnormal transfer setting used in this study. No additional data balancing techniques, such as oversampling, under-sampling, or synthetic augmentation, were applied in order to maintain the authenticity and clinical validity of the original data.

4.3. Implementation details

To ensure a robust evaluation of generalization, the dataset was split using an inter-patient protocol, with 80% of subjects for training/development and 20% of subjects for testing, ensuring no beats from the same patient appear in both sets. From the training/development portion, 20% was further reserved as a validation split for hyperparameter selection. The 10-fold cross-validation analysis reported in Section 5.6 was performed only within the 80% training/development portion using patient-level folds to assess training stability, while the independent 20% test set was kept fixed and was not included in any cross-validation fold. For the EAMCL loss (Section 3.4), the balancing coefficient was fixed to 0.1, selected on the validation split to maximize macro-F1; λ was constrained to [0,1] in implementation to maintain

Table 1

Class distribution of the processed ECG datasets after preprocessing and label mapping.

Dataset	N	S	V	F	Q	Abnormal	Total
MIT-BIH	90,585	2,780	7,2400	810	8,040	–	109,455
SVDB	162,213	12,194	9,936	23	79	–	184,445
INCART	153,591	1,958	20,000	219	6	–	175,774
PTB	4,046	–	–	–	–	10,506	14,552

stable optimization. Branch level training and hyperparameter sensitivity analyses used a maximum budget of 50 epochs, while the final fusion model was trained with validation-based early stopping. Adam was used with a learning rate of 1e-3, weight decay of 1e-4, and batch size of 128.

The training procedure was organized into branch-level training followed by fusion-stage optimization. During branch-level training, the low-level BiLSTM branch and the high-level CapsNet branch were trained independently to evaluate their standalone representational capacity, support ablation analysis, and guide hyperparameter selection. Both branches used the same maximum training budget of 50 epochs and the same optimization settings. The reported training times of approximately 30 s/epoch for the BiLSTM branch and 40 s/epoch for the CapsNet branch refer only to the computational wall-clock cost of one epoch for each standalone branch, rather than to different numbers of training epochs.

After branch-level evaluation, the complete dual-branch model was optimized through synchronized fusion-stage training. For each mini-batch, the BiLSTM branch and CapsNet branch were executed within the same forward pass, their projected representations were fused by the AMBAF module, and a single classification objective was used to update the trainable parameters through one backpropagation step. Thus, the fusion model did not use separate epoch schedules for the two branches. Fusion-stage training was controlled by validation-based early stopping and typically converged after approximately 30 epochs, as shown in the convergence analysis.

In the low-level branch, three convolutional layers with kernel sizes of 3, 5, and 7 were used to capture multi-scale ECG patterns. At 360 Hz, these correspond to short temporal windows of approximately 8 ms, 14 ms, and 19 ms, respectively, enabling the extraction of fine local waveform that are subsequently integrated by the BiLSTM for broader temporal context modeling. Meanwhile, the high-level branch uses a capsule hierarchy for high-level representation learning: 8 primary capsule types with capsule dimension $d_p = 16$, followed by 10 secondary capsules and 5 class capsules with capsule dimension $d_c = 32$. Enhanced dynamic routing is applied for $R = 5$ iterations in both the secondary and class capsule layers, and the capsule-attention module uses 2 heads. Dropout was applied to reduce overfitting ($p = 0.3$). Collectively, these settings promote stable training and robust hierarchical feature extraction.

4.4. Evaluation metrics

The performance of the proposed framework is assessed using accuracy, precision, recall, and F1-score as evaluation metrics, which are widely adopted in ECG classification studies due to their effectiveness in capturing model performance across diverse scenarios (Dalianis and Dalianis, 2018). Accuracy measures the overall correctness of predictions, providing a broad indicator of the model's reliability. Precision evaluates the proportion of correctly identified positive cases among all predicted positives, while recall quantifies the ability to detect all relevant instances within a class, particularly critical for addressing class imbalance in datasets like MIT-BIH. The F1-score, as the harmonic mean of precision and recall, offers a balanced assessment of performance on minority classes. These metrics were chosen to enable direct comparison with previous studies, ensuring the results align with established benchmarks and facilitating a comprehensive evaluation of the

proposed approach. The equations for these selected metrics are as follows:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (37)$$

$$Precision = \frac{TP}{(TP + FP)} \quad (38)$$

$$Recall = \frac{TP}{(TP + FN)} \quad (39)$$

$$F1 = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)} \quad (40)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

5. Results and Discussion

5.1. Overall performance comparison

The proposed model achieves highly competitive state-of-the-art performance when evaluated against several contemporary models. Table 2 summarizes the overall performance metrics, including accuracy, precision, recall, and F1-score. All comparisons are based on the MIT-BIH Arrhythmia Database under the AAMI five-class protocol (N/S/V/F/Q) with a patient-wise split (no subject overlap between training and test), and the results are presented as percentages (%). For methods reporting repeated-run or cross-validation results, the mean $\pm$ standard deviation is included; otherwise, only the reported single values are shown.

Table 2

Performance comparison of the proposed model and the counterparts.

Model	Year	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
HARDC (Islam et al., 2023)	2023	99.60	97.66	99.60	98.21
TCGAN (Xia et al., 2023)	2023	94.69	94.69	94.69	94.69
ECGformer (Akan et al., 2023)	2023	98	98	98	97
ACA-MA (Wang et al., 2023)	2023	99.38	99.38	97.02	98.18
MambaCapsule (Xu et al., 2024)	2024	99.54	96.76	89.01	93.50
ECGTransForm (El-Ghaish and Eldele, 2024)	2024	99.35 $\pm$ 0.16	95.47 $\pm$ 1.59	93.15 $\pm$ 1.16	94.26 $\pm$ 0.28
CBGM (Bai et al., 2024)	2024	99.41	99.15	99.10	97.80
CovHMM (Pan and Wu, 2025)	2025	94.11 $\pm$ 0.77	79.95 $\pm$ 3.61	89.12 $\pm$ 2.52	82.32 $\pm$ 1.80
C-BiNET (Febeena and Kurian, 2025)	2025	98.24	98.31	98.29	98.35
STL-HMFF (Talukder, 2025)	2025	99.30	99.31	99.30	99.30
Our	2026	99.55 $\pm$ 0.06	99.55 $\pm$ 0.06	99.55 $\pm$ 0.06	99.55 $\pm$ 0.06

The model attains 99.55% accuracy, precision, recall, and F1-score, reflecting consistently high performance across all evaluation metrics. In contrast, several previous models exhibit notable imbalances. For instance, MambaCapsule achieves a high accuracy of 99.54% but a recall of only 89.01% and an F1-score of 93.50%, indicating challenges in capturing minority-class instances. CovHMM demonstrates limitations with a precision of 79.95%, recall of 89.12%, and F1-score of 82.32%, suggesting issues in feature representation. ECGTransform shows occasional class confusion, as reflected by a precision of 95.47%, recall of 93.15%, and F1-score of 94.26%. Similarly, ECGformer, despite utilizing a sophisticated Transformer encoder, reaches about 98% overall accuracy but faces significant sensitivity gaps. ACA-MA, which employs multi-head self-attention on denoised signal segments, attains 99.38% accuracy and an overall sensitivity of 97.02%, indicating moderate recall limitations.

Several models employ advanced data augmentation or oversampling strategies to improve classification performance. TCGAN-augmented classifiers, although innovative in synthetic data generation, achieve only 94.69% accuracy, indicating that synthetic-data generation alone may not fully resolve class-imbalance-related challenges. HARDC applies Conditional GAN synthesis to generate additional samples for underrepresented classes and achieves an overall accuracy of 99.60%, but its precision and F1-score remain lower than those of the proposed model. C-BiNET leverages the ADASYN algorithm to generate synthetic minority samples and reports 98.24% accuracy. STL-HMFF combines SMOTE-Tomek Link oversampling with random oversampling to produce synthetic beats, attaining 99.30% accuracy. Despite these advanced approaches, the proposed model provides one of the most balanced overall profiles, achieving consistently high accuracy, precision, recall, and F1-score without relying on external synthetic beat generation.

This notable improvement underscores the robustness and effectiveness of the dual-branch model, which integrates temporal dynamics through BiLSTM with hierarchical features via Capsule Network, further enhanced by adaptive attention-based fusion. The architecture reduces bias by combining complementary information from low-level temporal and high-level capsule-based hierarchical domains, ensuring precise differentiation of arrhythmic from normal beats. This design minimizes false positives and false negatives, thereby enhancing the overall generalization capability of the model.

5.2. Per-class performance and imbalance mitigation

A detailed per-class analysis is crucial for understanding model behavior beyond the high-level summary provided by overall metrics, especially given the inherent class imbalance in arrhythmia datasets. This performance evaluation focuses on five AAMI classes (N, S, V, F, Q), unlike prior studies that often simplify the problem. Table 3 reports the per-class accuracy, precision, recall, and F1-score for the proposed method on the MIT-BIH dataset under the AAMI standard. These results clarify model behavior on both majority and minority beats and highlight the remaining difficulty of classes with limited samples.

The F1-score was selected for cross-method per-class comparison because it provides a balanced measure that captures both precision and recall in a single value, which is particularly important under class imbalance. Accordingly, Table 4 reports the per-class F1-score of each

Table 3
Per-class metrics on the MIT-BIH dataset (AAMI).

Class	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
N	99.83 ± 0.05	99.78 ± 0.06	99.83 ± 0.05	99.81 ± 0.05
S	94.42 ± 1.20	96.33 ± 1.05	94.42 ± 1.20	95.37 ± 1.10
V	98.76 ± 0.35	98.48 ± 0.40	98.76 ± 0.35	98.62 ± 0.35
F	89.51 ± 2.50	90.63 ± 2.30	89.51 ± 2.50	90.06 ± 2.30
Q	99.81 ± 0.12	99.88 ± 0.10	99.81 ± 0.12	99.84 ± 0.10

Table 4

Per-class F1-score of each method on the MIT-BIH dataset.

Method	N (%)	S (%)	V (%)	F (%)	Q (%)
HARDC (Islam et al., 2023)	97.57	97.21	97.38	91.2	97.58
TCGAN (Xia et al., 2023)	97.43	62.02	85.71	—	—
ECGformer (Akan et al., 2023)	99	73	93	66	98
ACA-MA (Wang et al., 2023)	96.96	93.92	97.99	98.01	—
MambaCapsule (Xu et al., 2024)	98.48	81.90	97.40	82.23	98.90
ECGTransform (El-Ghaish and Eldele, 2024)	99.41 ± 0.08	89.22 ± 0.22	96.69 ± 0.72	86.78 ± 1.22	99.18 ± 0.15
CovHMM (Pan and Wu, 2025)	96.48	67.87	82.42	68.84	98.48
Our	99.81 ± 0.05	95.37 ± 1.10	98.62 ± 0.35	90.06 ± 2.30	99.84 ± 0.10

method on the MIT-BIH dataset under the AAMI standard. Methods listed in Table 2 that report only aggregate metrics or do not adhere to the AAMI guidelines are omitted from this detailed per-class comparison. In addition, Fig. 4 presents the confusion matrix of the proposed model, showing the absolute sample counts for each true and predicted class.

Performance variations across classes, as shown in Table 4, are often attributable to the strategies used to handle imbalanced data. Several studies explicitly incorporate imbalance-mitigation strategies in ECG classification. HARDC employs a Conditional GAN to synthesize examples of underrepresented heartbeat classes, while ECGTransform integrates a Context-Aware Loss function to reweight minority classes during optimization. TCGAN uses a transformer-based GAN generator to augment scarce arrhythmia types. In contrast, methods such as MambaCapsule, CovHMM, ECGformer, and ACA-MA process imbalanced data directly without explicit mitigation strategies.

The current study employs the Enhanced Adaptive Margin Context Loss (EAMCL) to address class imbalance. As reported in Table 4, the proposed model achieves the highest F1-scores among the compared methods in three classes: N (99.81%), V (98.62%) and Q (99.84%). Results remain highly competitive for S (95.37%, compared to HARDC's 97.21% using Conditional GAN synthesis) and F (90.06%, compared to HARDC's 91.20% and ACA-MA's 98.01% with augmentation-based strategies). Unlike HARDC and ACA-MA, which depend on external data augmentation, EAMCL delivers robust performance across both majority and minority classes without requiring additional synthetic sampling.

5.3. Model generalizability

Mitigating the risk of overfitting is a persistent challenge in the development of deep learning models (Alzubaidi et al., 2023). To provide stronger evidence of generalization, cross-dataset transfer experiments were conducted using a transfer-learning protocol on PTB, INCART, and SVDB. Two adaptation strategies were evaluated: linear probing (freezing the feature extractor and training only the classifier head) and full-parameter fine-tuning. These two settings characterize feature transferability and model adaptability under domain shift. The corresponding results are reported in Table 5.

As shown in Table 5, linear probing achieves 93.74% on INCART (50 epochs), 89.23% on SVDB (50 epochs), and 90.10% on PTB (49 epochs). Full-parameter fine-tuning further improves performance, reaching 99.33% on INCART (13 epochs), 98.04% on SVDB (12 epochs), and 98.86% on PTB (13 epochs). Overall, the results indicate that the learned representations transfer effectively across datasets, while fine-tuning enables rapid and highly accurate adaptation under cross-

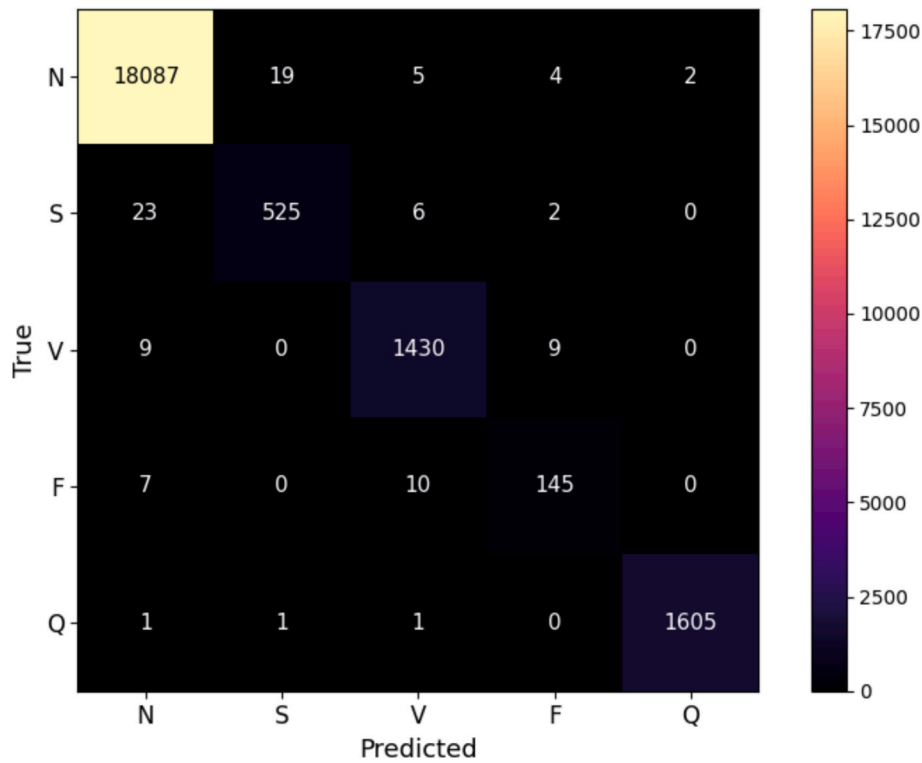


Fig. 4. Confusion matrix of the proposed model on the MIT-BIH test set under the AAMI five-class protocol.

Table 5

Cross-dataset transfer evaluation (linear probing vs. fine-tuning) on PTB, INCART, and SVDB.

Dataset	Stage	Epochs	Accuracy (%)	F1-score (%)
INCART	Linear Probe	50	93.74 ± 0.31	93.74 ± 0.33
INCART	Fine-Tune	13	99.33 ± 0.07	99.33 ± 0.08
SVDB	Linear Probe	50	89.23 ± 0.45	89.23 ± 0.47
SVDB	Fine-Tune	12	98.04 ± 0.16	98.04 ± 0.18
PTB	Linear Probe	49	90.10 ± 0.38	90.10 ± 0.40
PTB	Fine-Tune	13	98.86 ± 0.10	98.86 ± 0.11

dataset variability.

5.4. Comparative experiment: loss function evaluation

The Enhanced Adaptive Margin Context Loss (EAMCL) was comparatively evaluated against six baseline loss functions (cross-entropy, margin, focal, hinge, asymmetric, context) on high and low-level architectures (CapsNet and BiLSTM). Fig. 5 illustrates EAMCL's superior F1-score, a particularly reliable metric for class-imbalanced datasets, thus highlighting its strength in such scenarios. Notably, EAMCL achieved an improvement of over 2% in F1-score compared to all baseline losses. This superior performance demonstrates EAMCL's effectiveness in improving classification across all classes. This method achieves this by explicitly addressing the imbalance through dynamically adjusting the penalty imposed on misclassifications, preventing the

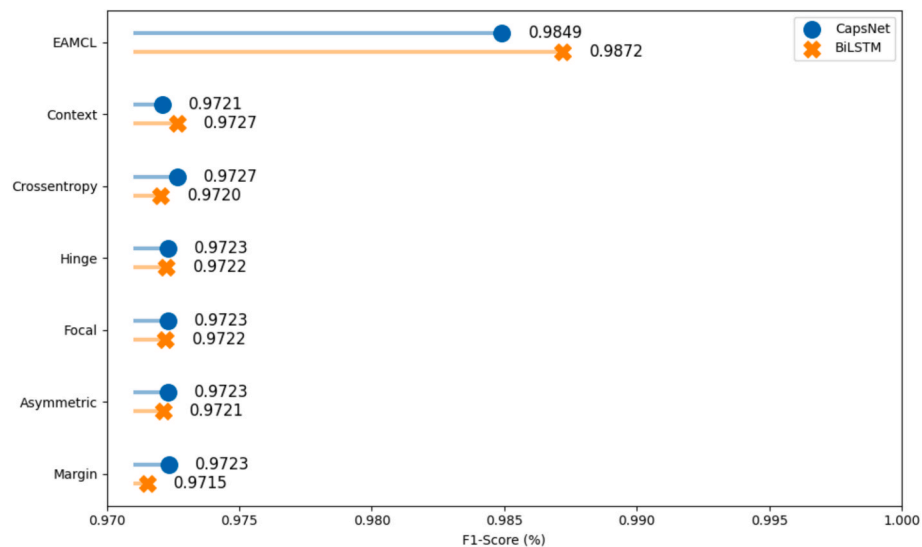


Fig. 5. Comparison of F1-Score for EAMCL and Baseline Loss Functions on CapsNet and BiLSTM Branch.

model from overfitting to the majority classes.

5.5. Ablation study: fusion strategies

To systematically assess the impact of each model component and fusion strategy, a comparative analysis was performed. This analysis evaluated the performance of the low-level BiLSTM branch, the high-level Capsule branch, the dual-branch fusion using both simple concatenation and bidirectional attention, and a static “No attention” fusion baseline. Table 6 summarizes the results, reported as overall classification accuracy on the MIT-BIH test set.

The ablation results highlight several key insights regarding the effectiveness of the dual-branch fusion and the necessity of an adaptive fusion mechanism:

- BiLSTM only and Capsule only branches individually achieve strong performance, demonstrating the discriminative power of both temporal and hierarchical features when used independently, with accuracies of 98.62% and 98.49%, respectively.
- The No attention variant fuses the projected and pooled branch features by simple averaging (an equal-weighted static mean) and shows a dramatic drop in performance (65.27%). This indicates that simply combining the branches with a fixed fusion rule, without any learnable or adaptive mechanism, can severely impair the model's ability to preserve and exploit complementary information; notably, this low value corresponds to the static-mean baseline rather than feature concatenation.
- The Fusion (Concatenation) variant combines features from the two branches through direct concatenation and achieves 98.18%, yet it does not outperform either single-branch model. This suggests that concatenation alone may introduce redundant or less informative components and does not reliably capture cross-branch interactions without adaptive selection/reweighting.
- The Fusion (Bidirectional) variant, employing adaptive multi-head bidirectional attention for dynamic fusion of branch features, attains the highest accuracy (99.55%). This confirms that a learnable, attention-based fusion strategy can optimally balance and exploit the strengths of both temporal and hierarchical representations, resulting in superior classification performance.

5.6. Stability analysis

Reliable training behavior is essential for clinical deployment. For clarity, the 10-fold cross-validation in this section was conducted only on the 80% training/development portion after the independent 20% test set had been fixed. The 80% training/development subjects were divided into 10 patient-level folds; in each fold, nine folds were used for training and one fold for validation to record stability indicators, including loss, attention statistics, attention entropy, and gradient norm. The independent test set was not used in this procedure. This section focuses on stability indicators of the dual-branch architecture, namely attention statistics, attention entropy, and gradient-norm monitoring, tracked over this 10-fold validation procedure. Because the bidirectional fusion layer aggregates information from both branches, stability is

Table 6
Ablation study results for fusion and branch variants.

Variant	Fusion mechanism	Test accuracy (%)
BiLSTM only	Single branch (BiLSTM)	98.62 ± 0.12%
Capsule only	Single branch (Capsule Network)	98.49 ± 0.05%
Fusion (No attention)	Feature projection + static-mean	65.27 ± 3.20%
Fusion (Concatenation)	Feature concatenation	98.18 ± 0.44%
Fusion (Bidirectional)	Adaptive bidirectional attention fusion	99.55 ± 0.06%

reported at the branch level (BiLSTM and CapsNet), where these indicators are most straightforward to quantify.

A. Low-Level Branch (BiLSTM) Attention and Stability.

As summarized in Table 7, the BiLSTM branch exhibits a low mean attention weight (~ 0.0053) with standard deviations in the 0.0299–0.0404 range, indicating occasional selective emphasis without persistent concentration. Because attention is normalized across many time steps, the mean value is expected to be small; robustness is more directly reflected by the entropy and spread. Attention entropy values consistently exceed 1.8 and peak at 2.5118, suggesting a broadly distributed attention pattern without collapse. This indicates that temporal evidence is aggregated from multiple segments rather than being dominated by a single peak, which reduces sensitivity to local perturbations (e.g., brief noise spikes or transient artifacts) and supports robust generalization across folds. Gradient norms range from 0.0896 to 3.2649, with the majority below 3, consistent with numerically stable updates during training.

B. High-Level Branch (CapsNet) Attention and Stability.

Table 8 reports the CapsNet branch statistics. Mean attention remains close to the theoretical uniform value for five capsules (~ 0.2000), with standard deviations up to 0.1289, indicating stable yet adaptable weighting. Attention entropy values remain near the theoretical optimum (≈ 1.6094), again showing no evidence of collapse. Entropy near the five-capsule optimum implies balanced utilization of multiple capsules rather than persistent domination by one capsule, which mitigates single-feature reliance and improves robustness to inter-patient variability and noise. Gradient norms fall between 0.0932 and 1.5893, supporting stable optimization dynamics.

Across folds, these attention statistics, entropy values, and gradient norms are consistent with stable training behavior. Fig. 6 visualizes fold-wise stability trends for both branches.

Fold-wise trends of (a) training loss, (b) attention spread (std), (c) attention entropy, and (d) gradient norm are shown for the BiLSTM and CapsNet branches (Fig. 6) and summarized numerically in Table 7 and Table 8. Gradient norms remain bounded across folds (BiLSTM: 0.0896–3.2649; CapsNet: 0.0932–1.5893), supporting stable optimization without attention collapse or gradient explosion. Building on this stability evidence, the next section reports targeted sensitivity analyses used to select the final training budget and key architectural hyperparameters for the BiLSTM and CapsNet branches.

5.7. Hyperparameter optimization and training stability

Targeted sensitivity analyses were conducted to balance model complexity, convergence speed, and generalization across both branches. Across folds, training dynamics plateaued well before the 50-epoch limit; therefore, 50 epochs was set as a conservative maximum budget for branch-level training and sensitivity analyses (see the branch

Table 7
BiLSTM branch attention and stability summary.

Fold	Loss	Attn. mean ± std	Attn. entropy	Grad norm
1	0.0581	0.0053 ± 0.0332	2.1162	2.7857
2	0.0614	0.0053 ± 0.0318	2.1597	3.2649
3	0.0552	0.0053 ± 0.0352	2.1044	0.2546
4	0.0564	0.0053 ± 0.0404	1.8099	1.3377
5	0.0605	0.0053 ± 0.0362	1.9559	0.1773
6	0.0536	0.0053 ± 0.0317	2.1508	0.1955
7	0.0605	0.0053 ± 0.0299	2.5118	0.6548
8	0.0516	0.0053 ± 0.0320	2.3626	0.0896
9	0.0602	0.0053 ± 0.0319	2.0901	2.8949
10	0.0568	0.0053 ± 0.0363	1.8281	3.0187

Table 8

Capsule network branch attention and stability summary.

Fold	Loss	Attn. mean $\pm$ std	Attn. entropy	Grad norm
1	0.072	0.2000 $\pm$ 0.0433	1.5873	0.1062
2	0.0663	0.2000 $\pm$ 0.0002	1.6094	1.5893
3	0.0801	0.2000 $\pm$ 0.0000	1.6094	0.0932
4	0.0713	0.2000 $\pm$ 0.0000	1.6094	0.1208
5	0.0715	0.2000 $\pm$ 0.0977	1.4938	0.1322
6	0.0727	0.2000 $\pm$ 0.1289	1.4392	0.1111
7	0.0697	0.2000 $\pm$ 0.0867	1.5221	0.4291
8	0.0729	0.2000 $\pm$ 0.0601	1.5691	0.5114
9	0.0722	0.2000 $\pm$ 0.0658	1.5604	0.3312
10	0.0711	0.2000 $\pm$ 0.0000	1.6094	0.3994

learning curves in Section 5.8(A–B)).

5.7.1. Sensitivity analysis of EAMCL hyperparameters

A validation based sensitivity analysis is conducted to justify the selection of α and $m_{\max}$ in the proposed EAMCL. The parameter α controls the sensitivity of the adaptive margin to class rarity, while $m_{\max}$ caps the maximum allowable margin. As shown in Fig. 7, different combinations of $\alpha \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ and $m_{\max} \in \{0.20, 0.35, 0.50, 0.65, 0.80\}$ were evaluated using macro-F1 score. The results obtained indicate that increasing $m_{\max}$ improves performance up to $m_{\max} = 0.50$, after which the macro-F1 score decreases. Similarly, $\alpha = 0.25$ provides the best balance between minority class emphasis and overall classification stability. The highest macro-F1 score is achieved at $\alpha = 0.25$ and $m_{\max} = 0.50$, confirming this combination as the most effective setting. Therefore, these values were adopted in the final EAMCL configuration.

5.7.2. Optimization of temporal feature extraction (BiLSTM)

BiLSTM sensitivity was assessed using two core parameters: hidden

size and stacked depth. Hidden units $\{64, 128, 256\}$ were evaluated. The 64-unit setting underfit long-range dependencies and reduced F1 by $\sim 1.2\%$ relative to the best configuration, while the 256-unit setting provided marginal training gains but showed validation overfitting and substantially higher FLOPs. The 128-unit setting achieved the best balance of accuracy and efficiency, reaching 98.62% (Fig. 8(a)).

Stacked depth was tested with 1, 2, and 3 BiLSTM layers. A single layer was insufficient for higher-level temporal abstraction, while three layers produced diminishing returns and less stable early training. A 2-layer BiLSTM with residual connections was therefore adopted, achieving 98.62% with lower gradient-variability proxy compared with deeper stacking (Fig. 8(b)).

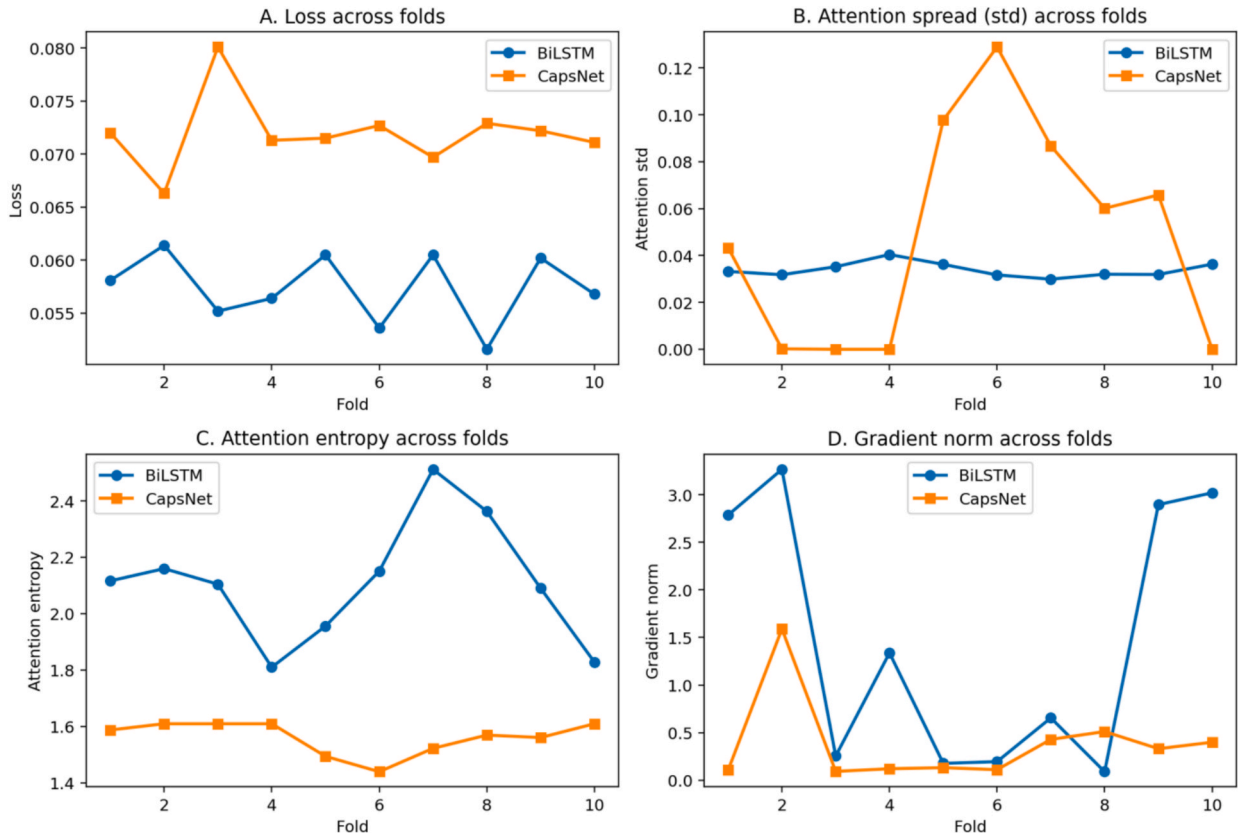
5.7.3. Optimization of hierarchical feature extraction (CapsNet)

CapsNet behavior is primarily governed by dynamic routing iterations (R) and capsule dimensionality. Routing iterations were evaluated for $R \in \{1, 3, 5, 7\}$. With $R = 1$, part-whole modeling was insufficient (Accuracy 97.45%). Performance improved at $R = 3$ and peaked at $R = 5$ (98.49%). Increasing to $R = 7$ yielded negligible benefit while increasing training latency by $\sim 18\%$, making $R = 5$ the most efficient choice (Fig. 9(a)).

Capsule dimensions were tested using Small ($d_p = 8, d_c = 16$), Medium ($d_p = 16, d_c = 32$), and Large ($d_p = 32, d_c = 64$) settings. The Small configuration reduced representational capacity, whereas the Large configuration increased model size by $\sim 2.5\times$ without improving validation performance, indicating over-parameterization. The Medium configuration ($d_p = 16, d_c = 32$) provided the best accuracy–complexity trade-off, achieving 98.49% (Fig. 9(b)).

5.8. Convergence analysis of model training

This section presents a detailed convergence analysis for each component of the framework. By examining the training and validation

**Fig. 6.** Stability across 10-fold cross-validation.

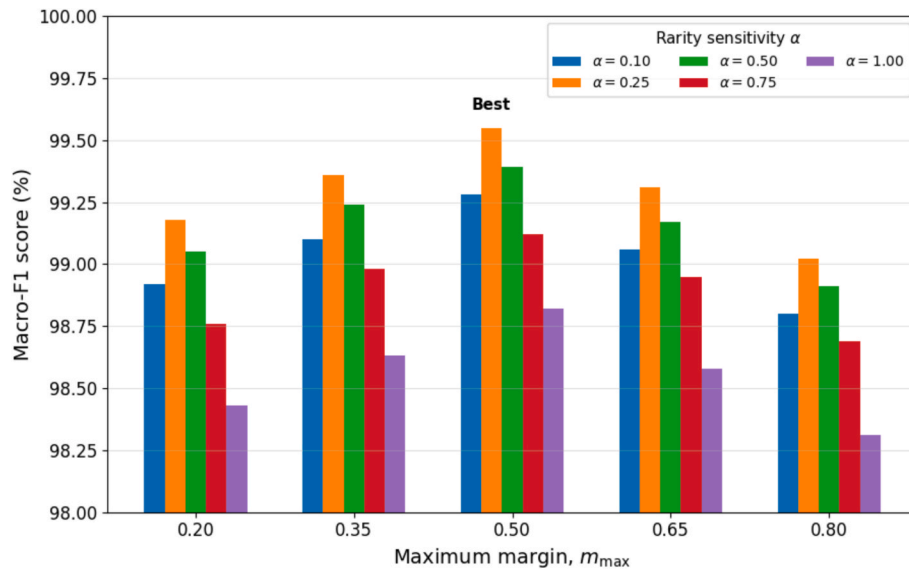


Fig. 7. Sensitivity analysis of EAMCL hyperparameters using macro-F1 score.

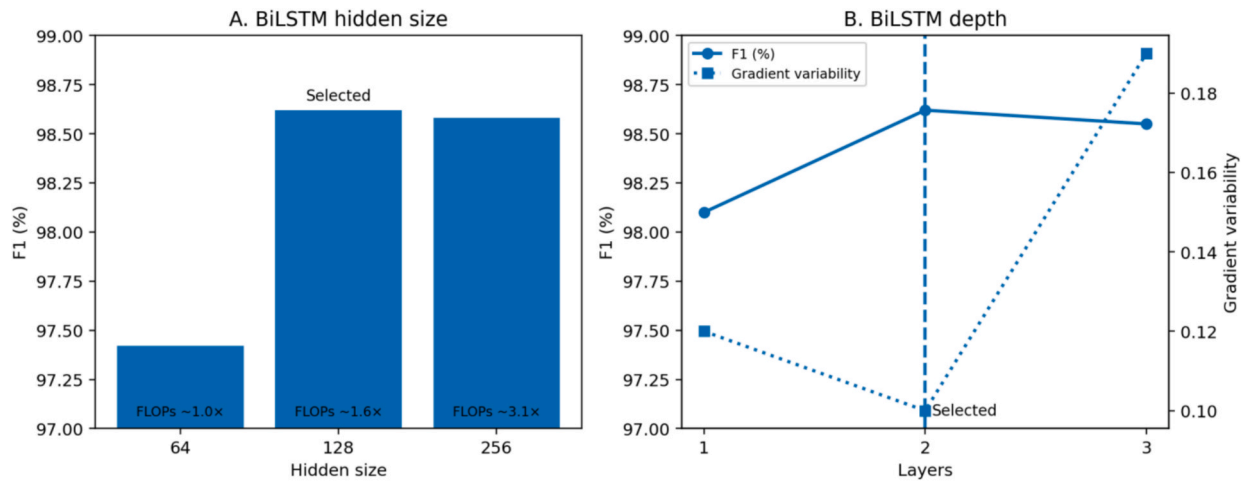


Fig. 8. BiLSTM sensitivity: hidden size and depth.

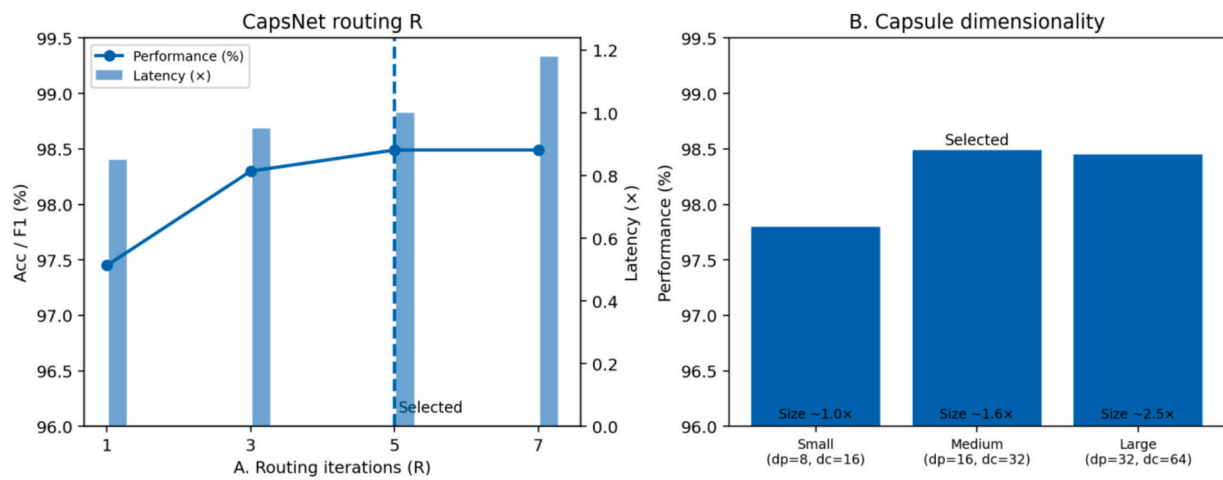


Fig. 9. CapsNet sensitivity: routing and capsule size.

loss and accuracy curves across epochs.

A. Convergence of the Low-Level Branch:

The convergence behavior of the low-level branch is illustrated in Fig. 10. Both the training and validation losses exhibit a steady decline during the early epochs, eventually plateauing below 0.06 after approximately 40 epochs. The validation accuracy rises rapidly to exceed 98% and stabilizes near 99%, mirroring the training accuracy throughout the process. This close alignment between the training and validation curves indicates strong generalization and a low risk of overfitting. Minor oscillations observed in the validation loss are anticipated, given the inherent variability present in ECG segments and class imbalance, but do not undermine the model's overall stability or effectiveness.

B. Convergence of the High-Level CapsNet Branch:

The training dynamics of the high-level CapsNet branch are presented in Fig. 11. The training loss decreases sharply within the initial 20 epochs and then levels off, with the validation loss showing a parallel trend. The accuracy for both training and validation datasets exceeds 98.5% by epoch 20, and the gap between the two curves remains consistently under 1%. This behavior highlights the robust learning capability of the branch, as the hierarchical capsule representations combined with multi-head self-attention effectively capture key hierarchical features in the ECG data, all while avoiding significant overfitting.

C. Convergence of the Fusion Model:

The training trajectory for the full model with Adaptive Multi-head Bi-Directional Attention Fusion is shown in Fig. 12. Both training and validation losses rapidly decrease and converge to low values (≈ 0.02), while validation accuracy exceeds 99% early in training. Minimal divergence is observed between training and validation curves, indicating limited overfitting as accuracy approaches saturation. Small fluctuations in training loss may occur due to the adaptive reweighting behavior of the attention fusion module as it balances complementary features from both branches. Early stopping is applied to the fusion model based on validation performance, and training typically terminates at approximately 30 epochs.

The convergence curves for each component and the fused model confirm fast, stable, and reliable learning dynamics. The minimal overfitting evidenced by the close tracking of training and validation metrics reinforces the effectiveness of the proposed architectural and training strategies. These findings, which are consistent with the broader stability analysis, underscore the practical robustness and reliability of the model for clinical ECG classification tasks.

5.9. Model complexity

Reporting model complexity is essential for future researchers to make meaningful comparisons and track the evolution of AI models. However, this practice is often overlooked in the current literature.

Specifically, several benchmark methods summarized in Table 2 do not report model parameter counts, model size, or FLOPs. Furthermore, some of these methods do not provide public code repositories, which prevents a direct complexity-based comparison. Therefore, to ensure a fair and verifiable comparison, Table 9 includes only counterpart methods for which complexity values were either reported in the original studies or obtained from publicly available source code. To better demonstrate the performance-complexity trade-off, Table 9 also reports the F1-score of each method.

As shown in Table 9, the proposed model provides a favorable balance between classification performance and computational efficiency. Although ECGformer, CovHMM, and TCGAN have fewer parameters or smaller model sizes, their reported F1-scores are lower than that of the proposed model. Compared with ECGTransForm, the proposed model achieves lower FLOPs while maintaining stronger classification performance. These results indicate that the proposed model does not simply rely on excessive computational scale, rather its performance gain is mainly attributed to the effective integration of complementary temporal and hierarchical representations through the adaptive bidirectional fusion mechanism.

To further clarify the source of the proposed model complexity, the branch-level parameter distribution was examined. The low-level BiLSTM branch is lightweight, containing only 62.3 K parameters, whereas the high-level CapsNet branch accounts for most of the model capacity, with approximately 3.7 M parameters. The fusion module contains only 64.9 K parameters. Although only the fusion module is trainable during the final fusion-stage training, the total parameter count is reported because all components are loaded and used during inference. This distribution shows that the additional complexity of the proposed architecture mainly originates from the CapsNet branch, which includes capsule transformation matrices, hierarchical capsule layers, dynamic routing operations, and capsule-level attention. These components are designed to preserve high-level morphological relationships in ECG waveforms. Therefore, the model's complexity is mainly introduced by the hierarchical morphology modeling component rather than the temporal BiLSTM branch or the fusion module.

Collectively, the comparison in Table 9 demonstrates that the proposed model achieves strong predictive performance with moderate computational complexity. This confirms its practical suitability for real-world clinical ECG analysis deployment, particularly on GPU-accelerated diagnostic platforms.

5.10. Model bias and explainability

Reducing model bias and enhancing explainability are critical for creating trustworthy AI models. This section explores potential sources of bias in the proposed approach and the strategies used to address them.

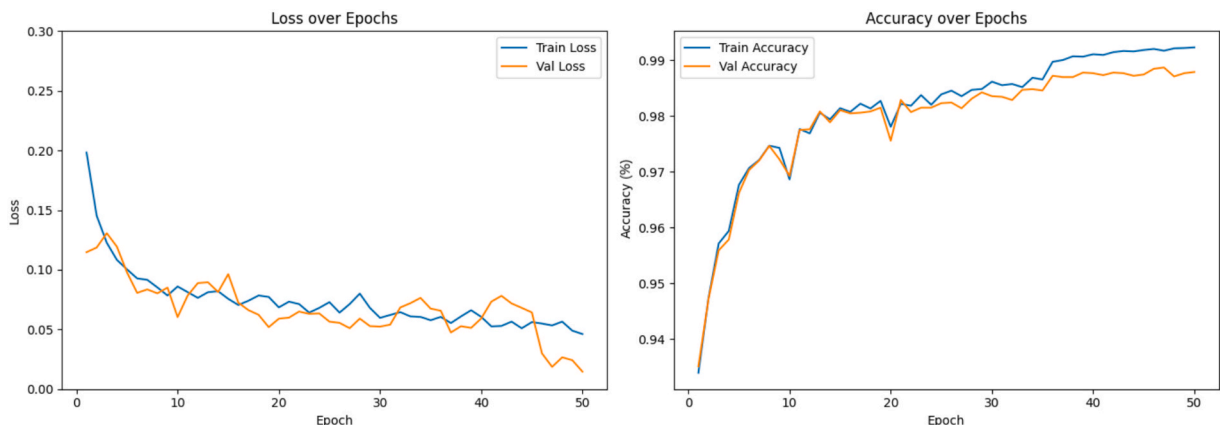


Fig. 10. BiLSTM learning curves (loss and accuracy vs. epoch).

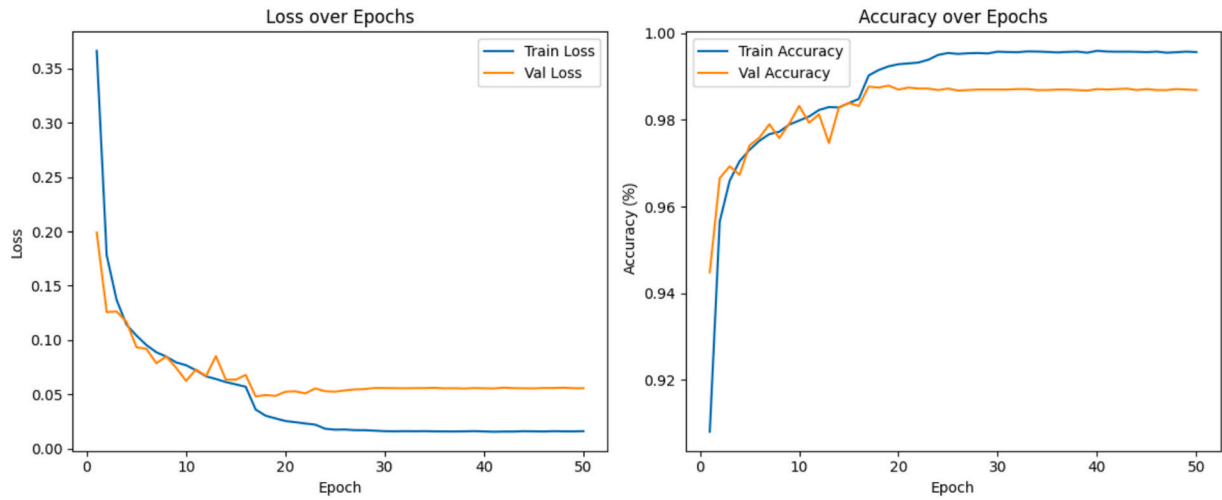


Fig. 11. CapsNet learning curves (loss and accuracy vs. epoch).

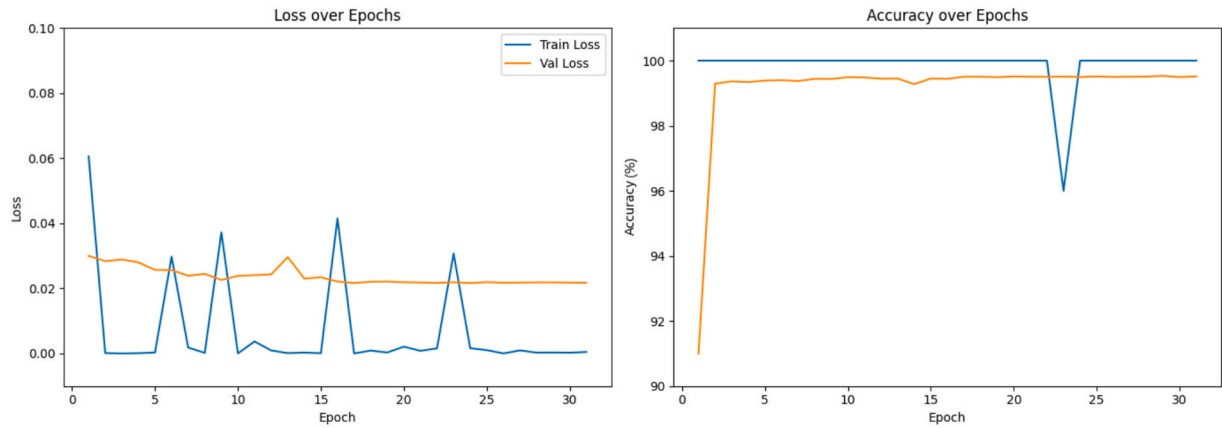


Fig. 12. Fusion Model learning curves (loss and accuracy vs. epoch).

Table 9

Performance and model complexity comparison of the proposed model and selected counterpart methods.

Model	F1-score (%)	Total Parameters (M)	Model Size (MB)	FLOPs (M)
ECGformer (Akan et al., 2023)	97	0.036	0.15	55
ECGTransForm (El-Ghaish and Eldele, 2024)	94.26 ± 0.28	0.63	2.56	86.36
CovHMM (Pan and Wu, 2025)	82.32 ± 1.80	0.061	0.33	16
TCGAN (Xia et al., 2023)	94.69	0.79	3.21	10.13
Our	99.55 ± 0.06	3.78	15.61	50.74

- Dataset limitations can introduce bias, as highlighted by Albahri (Albahri et al., 2023). The datasets utilized in this study are imbalanced and incorporate multi-class samples, which could lead the model to favor labels with more plentiful training examples. To address this, EAMCL was implemented to equilibrate positive and negative samples in the multi-class classification task, as described in Section 3.4.
- The selection of evaluation metrics, accuracy, precision, recall, and F1-score, can impact fairness, as these metrics guide the selection of hyperparameters and trained models.

- An early stopping mechanism was employed during training to further reduce overfitting and improve model generalization, helping to prevent bias towards the training data.
- The proposed dual-branch architecture reduces bias by integrating complementary features from low-level temporal and high-level hierarchical domains, thus minimizing reliance on potentially biased features and enhancing the overall generalization capability of the model.

In terms of explainability, an interpretable signal is provided at the fusion stage by quantifying the relative contribution of each branch to the final decision. Specifically, branch contribution weights are computed post-hoc from the normalized mean absolute activations of the low-level (BiLSTM) and high-level (Capsule) branch representations entering the bidirectional fusion module, yielding two nonnegative weights that sum to one per test sample. As shown in Fig. 13, the mean $\pm$ standard deviation contribution on the MIT-BIH test set is 0.710 ± 0.044 for the low-level branch and 0.290 ± 0.044 for the high-level branch, indicating that temporal dynamics contribute more strongly on average while hierarchical morphological cues provide complementary support.

5.11. Noise sensitivity and statistical significance

To evaluate sensitivity to measurement artifacts, additive white Gaussian noise (AWGN) was injected into the test signals at signal-to-

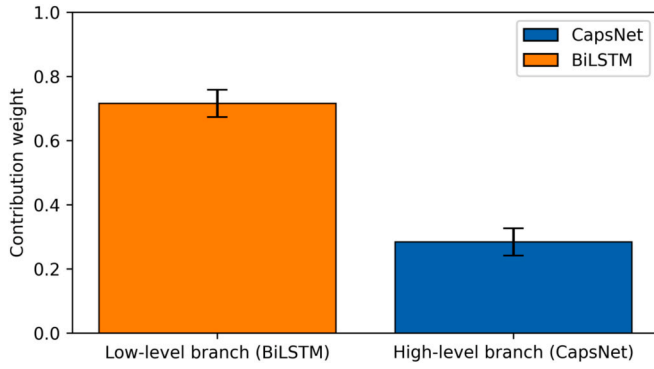


Fig. 13. Branch contribution weights of the bidirectional fusion module on the MIT-BIH test set.

noise ratios (SNRs) of 5 dB and 0 dB. Since the proposed model was not trained with noise-aware augmentation or a dedicated denoising front-end, this experiment is presented as a noise sensitivity analysis rather than as evidence of noise robustness. Performance metrics for the low-level branch (BiLSTM), high-level branch (CapsNet), and fusion AMBAF + EAMCL model are illustrated in Fig. 14.

To further contextualize the degradation under noisy conditions, two counterpart methods were evaluated under the same AWGN testing setting. The corresponding F1-score comparison is reported in Table 10. All models experience substantial degradation under AWGN corruption, confirming that severe noise remains challenging for ECG classifiers that are not explicitly optimized for noise robustness. The proposed model achieves the highest F1-score under clean and 5 dB conditions, while remaining comparable to the selected counterparts under 0 dB noise. These findings clarify that extreme-noise robustness remains a limitation and should be addressed in future work by investigating noise-aware training, SNR-adaptive fusion, and dedicated ECG denoising front-ends to improve performance under heavily corrupted signals.

Statistical Significance: Analysis on the clean test set ($N = 21,891$) confirms the significance of these gains. The 95% Wilson confidence intervals for accuracy are:

- low-level [98.46, 98.77]
- high-level [98.32, 98.64]
- combined [99.54, 99.70]

The fusion model yields a significant improvement over both the low-level and high-level branches under clean test conditions. With non-

Table 10

Noise sensitivity comparison under AWGN corruption.

Model	Clean F1-score (%)	F1-score at 5 dB (%)	F1-score at 0 dB (%)
ECGTransForm (El-Ghaish and Eldele, 2024)	94.26	29.55	22.89
TCGAN (Xia et al., 2023)	94.69	32.10	26.09
Our	99.55	32.46	25.46

overlapping confidence intervals for accuracy, the performance advantage provided by the AMBAF + EAMCL fusion strategy is statistically robust.

6. Conclusion and future work

This study proposes a robust ECG arrhythmia classification framework that combines a multi-scale CNN-SE-BiLSTM branch for temporal dynamics with a hierarchical Capsule Network branch enhanced by multi-head self-attention for morphology-aware representation learning. These complementary representations are integrated through Adaptive Multi-head Bi-Directional Attention Fusion (AMBAF), which explicitly captures cross-level interactions instead of relying on static fusion. Class imbalance is addressed through Enhanced Adaptive Margin Context Loss (EAMCL), which improves minority-class separability via class-aware adaptive margins without oversampling-based augmentation. In addition, fusion-stage branch contribution analysis provides an interpretable indicator of how temporal and hierarchical representations contribute to each decision.

Experimental validation using the MIT-BIH benchmark (AAMI five-class, inter-patient split) achieved $99.55 \pm 0.06\%$ accuracy, precision, recall, and F1-score, confirming strong and balanced performance across classes. Cross-dataset transfer on PTB, INCART, and SVDB further supported generalizability under domain shift, with full fine-tuning accuracies of 98.86%, 99.33%, and 98.04%, respectively. Robustness to acquisition artifacts was assessed via additive white Gaussian noise (AWGN) injection at SNR levels of 5 dB and 0 dB, showing that performance degrades as corruption increases and that macro-F1 can drop sharply under low-SNR conditions. Statistical significance analysis on the clean test set further confirmed the reliability of the observed gains, with the fused model achieving a higher accuracy confidence interval than either branch alone.

Several limitations remain. Robustness under very low SNR is still limited, indicating that fusion can be affected when one branch becomes unreliable under severe corruption, and the current robustness evaluation focuses on AWGN rather than other clinically common

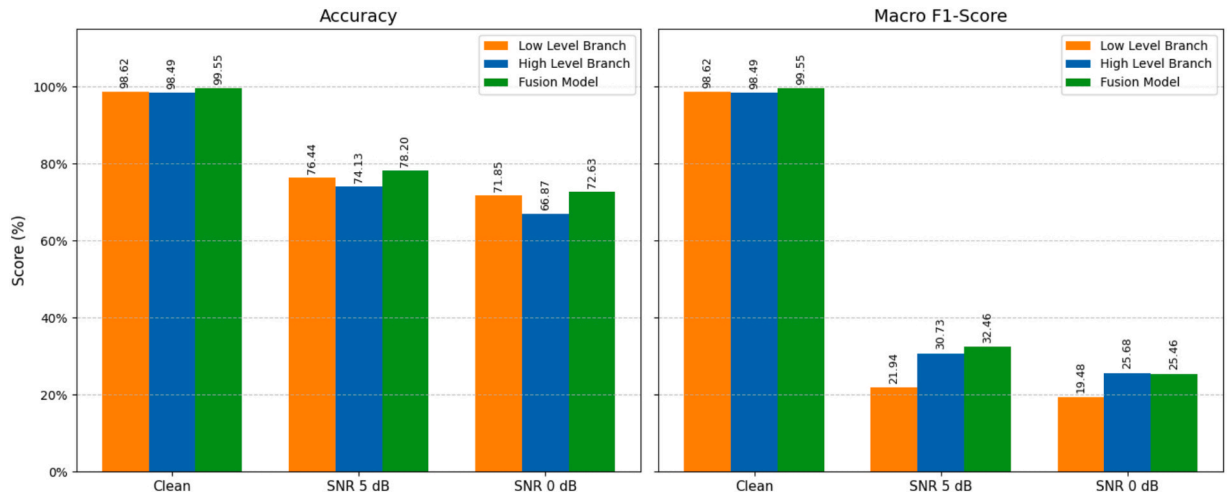


Fig. 14. Noise sensitivity across varying signal-to-noise ratios (SNR).

nonstationary artifacts (e.g., baseline wander, electromyography noise, and motion artifacts). Future work will therefore prioritize (i) noise-aware training (multi-SNR augmentation and curriculum noise schedules), (ii) signal-quality-guided fusion to adaptively gate or reweight branches based on estimated signal quality, and (iii) integration of front-end denoising or robustness regularization to stabilize performance in extreme acquisition conditions. Broader validation on additional datasets and acquisition settings, as well as efficiency improvements (e.g., pruning, quantization, and architecture refinement) to support deployment on resource-constrained platforms, will also be pursued.

CRedit authorship contribution statement

Basheer A. Hassoon: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Shengwu Xiong:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Mushtaq A. Hasson:** Data curation, Software, Validation, Formal analysis, Writing – review & editing. **Ridwan Salahudeen:** Methodology, Validation, Formal analysis, Writing – review & editing. **Aminu Onimisi Abdulsalami:** Methodology, Investigation, Writing – review & editing. **Tauqeer Khan:** Methodology, Investigation, Formal analysis, Visualization, Writing – review & editing.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics statement

This study used only publicly available, de-identified benchmark ECG databases and did not involve direct recruitment, intervention, or contact with human participants by the authors. Accordingly, no additional institutional ethical approval or informed consent was required for this retrospective computational analysis.

Data and Code Availability

The datasets used in this study are publicly available benchmark ECG databases (MIT-BIH (Moody and Mark, 2001), PTB (Bousseljot et al., 1995), INCART (Yakushenko, 2008), and SVDB (Mark et al., 1990). The implementation, training scripts, and evaluation pipeline are planned to be made publicly available on GitHub in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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